Demian Kong

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AI Agents & LLM Systems · Scientific AI · Reliable Execution

PROFILE

Bioinformatics M.S. graduate building reliable AI systems for scientific data and tool-driven work. Experience spans natural-language database access, verified Text-to-SQL, cross-database integration, and stateful agent runtimes. Current work focuses on inspectable planning, execution, review, and recovery.

EDUCATION

University of Chinese Academy of Sciences (UCAS) M.S. Bioinformatics · 2023–2026 · graduated Jun 2026

Thesis: *Development of Bioinformatics Data Retrieval Technology and Configurable Agent Framework Based on Large Language Models.*

Jilin University B.S. Pharmacy · 2019–2023

RESEARCH EXPERIENCE

National Genomics Data Center (NGDC), CNCB Graduate Researcher · 2023–Jun 2026

- Built conversational retrieval agents for genomic and epigenomic databases; designed multi-stage Text-to-SQL and interactive data-exploration workflows for EWAS AI Assistant.
- Contributed multi-omics pipeline and chatbot components to iDog; curated and integrated biomarker data from more than 20 public resources for BioKA.

Jilin University Undergraduate research · 2021–2023

- Investigated MKP5-related hepatic-fibrosis mechanisms and conducted organic synthesis and single-crystal X-ray crystallography.

SELECTED PROJECTS

Chrona · Current developer Repository · v0.3.1

- Local-first AI work executor for Task → Plan → Schedule → Execute → Review/Recover, with approval boundaries, persistent state, tool traces, and recovery. Current public release: v0.3.1 alpha.

Dingent · Founder and Lead Architect 2024–Jun 2026 · Preprint

- Built a configurable framework with MCP-based tools, graph-compiled workflows, checkpointed execution, and cross-database evidence integration; integrated it across eight CNCB data resources during graduate work.
- Evaluated under a 100-concurrent-user workload with approximately 400 ms median framework-side latency. Transferred the project to the group at graduation; now provide technical support only.

EWAS AI Assistant · Developer 2024–Jun 2026 · BioAgent

- Built entity retrieval, schema-constrained Text-to-SQL, execution-based verification/correction, synthesis, and visualization. On 30 expert-reviewed questions with reference SQL: 76.7% strict and 83.3% fully-or-partially-correct accuracy.

> Project figures above are project/thesis evaluation evidence, not journal-reported metrics.

SELECTED PUBLICATIONS

1. **Kong D**, Bei S, Wu Y, Tang B, Zhao W. "Dingent: An Easily Deployable Database Retrieval and Integration Agent framework." *bioRxiv preprint*, 2026. doi:10.64898/2026.03.17.712026.
2. Yang F†, Xiong Z†, Zong W†, **Kong D†**, et al. "EWAS Open Platform 2026." *Nucleic Acids Research*, 2026;54(D1):D1061–D1068. doi:10.1093/nar/gkaf1155.
3. Liu Y†, Wang Y†, Sun J†, **Kong D†**, Zhou B†, et al. "iDog: a multi-omics resource for canids study." *Nucleic Acids Research*, 2025;53(D1):D1039–D1046.
4. Wang Y, **Kong D**, et al. "BioKA: a curated and integrated biomarker knowledgebase for animals." *Nucleic Acids Research*, 2024;52(D1).
5. CNCB–NGDC Members and Partners. "Database resources of the National Genomics Data Center, China National Center for Bioinformation in 2026." *Nucleic Acids Research*, 2026;54(D1):D28–D47. doi:10.1093/nar/gkaf1172. **Kong D** is listed among consortium members.

† Equal contribution / joint first author.

TECHNICAL SKILLS & RECOGNITION

Programming & systems: Python, TypeScript/JavaScript, SQL, Bash, Linux, Docker, Git · **AI & applications:** PyTorch, LangChain/LangGraph, MCP, FastAPI, React, Next.js, Vue · **Languages:** Chinese (native), English (IELTS 7.0) · **Recognition:** Provincial Undergraduate Innovation and Entrepreneurship Training Program, Jilin University